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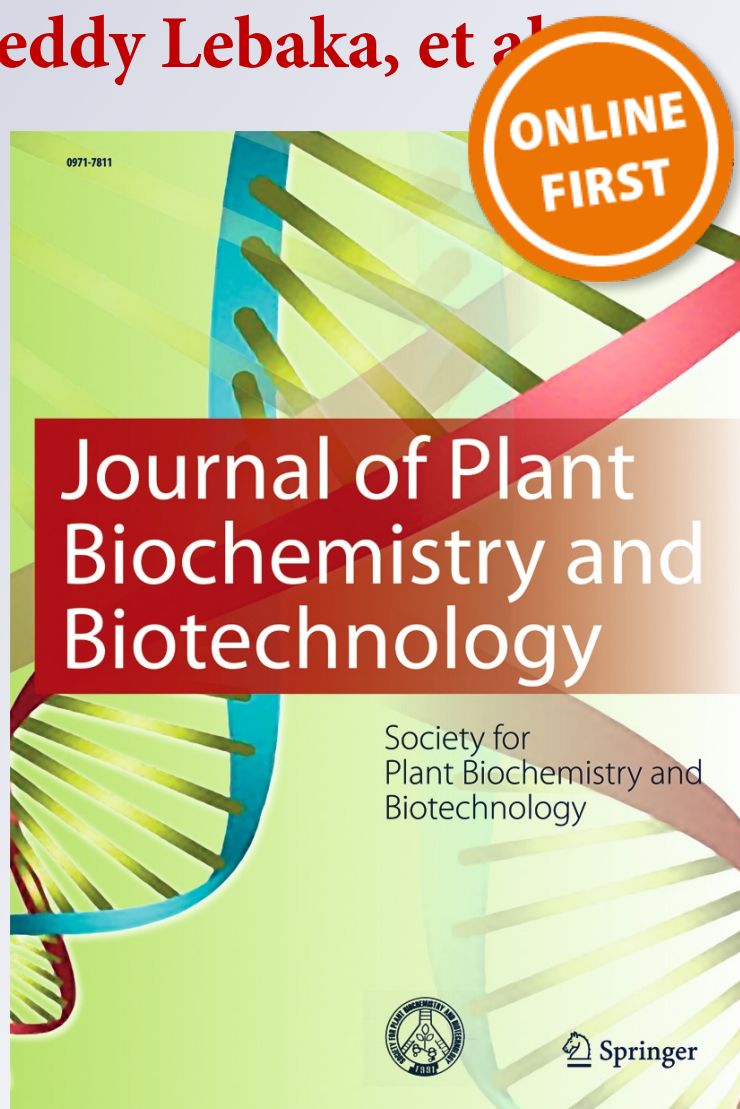
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Molecular cloning, characterization and three-dimensional structure prediction of Lipoxygenase from Finger millet [*Eleusine coracana* (L.) Gaertn.] germinating seedlings

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Abstract Lipoxygenases (LOXs) are functionally diverse class of non-heme iron containing dioxygenases. They involve in multiple physiological processes in plants and animals. In current study a full length cDNA of LOX has been cloned from finger millet germinating seedlings using PCR and RACE PCR methods. The ORF (2661 bp) encodes 887 amino acid residues, with an approximate molecular weight of 100.8 kDa and an isoelectric point of 5.90. The sequence exhibited significant similarity to already known LOXs. It comprises of a putative PLAT domain and an iron-binding catalytic domain. Sequence analysis demonstrated that this LOX is most probably a type 9-LOX, and shared highest identity (85 %) with the *Setaria italica* 9-LOX. TargetP predicted *EcLOX* as a cytoplasmic protein. It has an optimum pH of about 6.4 and increased lipid peroxidation levels were observed on day 5 of germination. Semi-quantitative RT-PCR analysis showed that *EcLOX* gene expression was preferentially higher on day 4 after germination, and in growing leaf and root tissues. Its expression could be induced by Methyl Jasmonate (MeJA), but not by Salicylic acid (SA). The 3-dimensional structure of the *EcLOX* was modeled using in silico

approach and Molecular dynamics (MD) simulations were performed. Further, to understand the protein active site, docking studies were performed with substrates and inhibitors. In summary, *EcLOX* protein was characterized using various molecular, biochemical and in silico methods which improves our understanding of these fascinating enzymes.

Keywords Finger millet · MD simulations · Molecular Docking · Molecular modeling · Plant lipoxygenases

Abbreviations

<i>EcLOX</i>	<i>Eleusine coracana</i> lipoxygenase
DOPE score	Discrete Optimized Protein Energy
MDA	Malondialdehyde
RACE PCR	Rapid Amplification of cDNA Ends Polymerization Chain Reaction
MEGA	Molecular Evolutionary Genetics Analysis

Introduction

Lipoxygenases (LOXs; E.C. 1. 13.11.12) are iron containing non-heme, non-sulfur oxidoreductases which catalyze the polyunsaturated fatty acids (PUFAs) by insertion of molecular O₂ into 1, 4- *cis, cis* -pentadiene structure (Andreou and Feussner 2009). This family proteins are widely distributed in mammals, amphibians, plants, fungi, prokaryotes and yeast (Joo and Oh 2012), demonstrating the biological and evolutionary importance of these enzymes in nature. Linoleic (18:2) and Linolenic (18:3) acids are the most copious fatty acids in the lipid fraction of plant cell membranes, these are the major substrates for LOXs (Vick 1993), and in plants these substrates are oxygenated at C-9 or C-13 position to produce 9 (S)-HPOD (T) E and 13 (S)-(9Z,11E)

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HPOD (T) E by 9- and 13-LOXs, respectively (Cho et al. 2011). The produced 9 (S)-HPOD (T) E and 13 (S)-(9Z,11E) HPOD (T) E are rapidly converted into a collective group of final compounds called oxylipins (Porta and Rocha-Sosa 2002). In plants, according to their regio-specificity 9- and 13-LOXs are the two groups of LOXs present, which specifically produce 9- and 13-hydroperoxy fatty acids (Feussner and Wasternack 2002). However, sometimes this classification encounters some difficulties because non-traditional LOXs yield a mixture of 9- and 13-hydroperoxy fatty acids upon catalysis (Liu and Han 2010). Earlier, LOXs were separated into two gene subfamilies type-1 and type-2, according to the absence (type-1) and presence (type-2) of an expected chloroplast transit peptide in their N-terminal side of proteins (Peng et al. 1994). In animals, LOX pathway leads to production of physiologically active compounds collectively known as leukotrienes and lipoxins. Their physiological functions have been described precisely by several groups (Funk 1996; Krieg and Fürstenberger 2014; Horn et al. 2014; Adel et al. 2015). The role of plant LOXs is more complex as they involve in a number of functions (Porta and Rocha-Sosa 2002). The metabolism of 13-hydroperoxides is mostly studied in plants because 13-hydroperoxy derivatives of LOXs are the precursors of Jasmonic acid. Jasmonates play a crucial role in plant developmental events such as pollen maturation, anther dehiscence, tuberisation and defence (Santino et al. 2005). The fate of 9-hydroperoxide derivatives has been so far less investigated. Recent report states that 9-LOX products are vital for lipid peroxidation during plant hypersensitive reactions, defense (Rusterucci et al. 1999), and implicated in different plant developmental processes like tuber growth, (Kolomiets et al. 2001) and late root development (Vellosillo et al. 2007). Nonenal or nonadienal (C9 aldehydes) is the major constituent of flavors and aromas of many fruits, converted from the 9-HPOD and 9-HPOT Hydroperoxide Lyase (HPL) branch pathway (Matsui 2006). These volatile substances produced via LOX pathway have great applications in food industry as they play an important role in maintaining food quality and the aroma of food grains (Eskin et al. 1977). Several 9-LOXs were cloned from various monocot species mainly from cereals such as *Zea mays*, *Oryza sativa*, *Hordeum vulgare* and *Triticum spp* (Park et al. 2010; Zhou et al. 2014; van Mechelen et al. 1995; Feng et al. 2010).

Spermatophytes life cycle starts with germination, lipid bodies are degraded during lipid mobilization in seeds by the new set of proteins, among these proteins LOXs play an important role in germination (Liavonchanka and Feussner 2006). Multiple isoforms of seed LOXs were identified in cereal crops, two from germinating barley and three from germinating grains (De Simone et al. 2010; Ohta et al. 1986; Garbe et al. 2006). To improve our knowledge on the role of LOXs during germination in cereals, we aimed to analyze germinating seedlings of finger millet. Here, we have cloned LOX gene from finger millet germinating seedlings. The temporal, tissue expression and hormonal regulation of the LOX gene was analyzed during

germination. The analysis suggested that this protein is a 9-LOX and is active during finger millet seed germination.

The information available regarding 3-dimensional structure of the plant LOXs is less. Methods like NMR analysis and X-ray crystallography are difficult to perform in determining a protein structure. So, *in silico* protein models generated using homology modeling provide a solid basis for exploring the protein architecture, predicting protein function, identifying the active site residues, structure-based drug designing, and studying the catalytic mechanisms. The composition of amino acid sequences of both plant and animal LOXs are different to a certain extent; however the 3-dimensional protein structures display a high degree of similarity (Kuhn et al. 2005). Most of our awareness on the physiological role and structural characteristics of LOXs were initiated from studies on the soybean LOX-1 (Shibata et al. 1988). Presently, X-ray crystal structure data of LOXs and enzyme–ligand complexes from few other plant sources is available (Ivanov et al. 2010). However, adequate amount of the structural data, particularly on the 3-dimensional structure of plant LOXs, and information about hot spot residues at the active site are very essential for understanding their mechanism of action in various physiological and pathological conditions. In this regard, homology modeling was performed to gain insights into the 3-dimensional structure of the *EcLOX*, and validated using various online tools and the time dependent behavior of the modeled protein was checked by performing MD simulations. Further, to explore the active site of the protein, docking studies were performed with substrates and inhibitors, and the results were reported.

Finger millet (*Eleusine coracana* L.) or Ragi is one of the popular millet cultivated in India. In southern region of India, Ragi is the chief food grain of the rural population (Ravindran 1991). Prolamins constitute a major fraction of finger millet proteins; lipid fraction contains 46–62 % oleic acid, 8–27 % linoleic acid, 20–35 % palmitic acid, and traces of linolenic acid (Mahadevappa and Raina 1978). Phenolic compounds from the millet seed coat displayed strong inhibition towards a glucosidase and pancreatic amylase (Shobana et al. 2009). Regular ingestion of entire grain cereals and their products can defend against the risk of cardiovascular diseases, type II diabetes, gastrointestinal cancers and a variety of other disorders (Devi et al. 2011). Finger millet is exceptionally rich in calcium (344 mg %) and potassium (408 mg %) compared to other cereals and millets and also contains 283 mg % of phosphorus, and 3.9 mg % of iron (Gopalan et al. 1971).

Materials and methods

Chemicals

All the necessary chemicals were purchased from Merck (Vadhani Industrial Estate, Mumbai, India), and Molecular

biology chemicals were procured from Takara (Takara Ostu, Shiga, Japan). RACE PCR kits were bought from Roche (Basel, Switzerland), and Methyl Jasmonate, Salicylic acid, and Linoleic acid were obtained from Sigma–Aldrich (St. Louis, MO USA).

Plant material and growth conditions

Finger millet seeds were obtained from Regional Agricultural Research station, Vizianagaram, Andhra Pradesh, India. Finger millet seeds (Sri Chaitanya VR-847) are surface sterilized with 0.1 % (w/v) HgCl₂ for 10 min; they are then imbibed in deionized water for 10 to 12 h and were allowed to germinate at 27 °C in the dark. After germination, the seedlings were spread on tissue paper and germination of the seeds was carried out at room temperature (15/9 h day/night) up to 7 continuous days.

Enzyme activity

The frozen plant material of the finger millet germinating seedlings was blended into a fine powder. One gram of powdered plant material was suspended in 10 times their mass of 50 mM phosphate buffer at pH 6.4 containing 1 mM Phenylmethylsulfonyl fluoride (PMSF). The resulting homogenate was then centrifuged for 10 min at 12,000g at 4 °C. The subsequent supernatant was assayed for LOX activity. Activity was measured as previously described using Shimadzu UV–VIS spectrophotometer, at 25 °C by monitoring the increase in absorbance over a period of time at 234 nm (Reddanna et al. 1990). The typical reaction mixture contains 2.8 ml of 50 mM sodium phosphate buffer at pH 6.4, the appropriate volume of the enzyme (10–100 µl) and the reaction was initiated by addition of substrate to the reaction mixture making sure to maintain 250 µM for linoleic acid in the total volume. One unit of enzyme activity is defined as the amount of micro moles of hydroperoxide formed per minute. The crude enzyme samples inactivated by heat treatment at 100 °C for 10 min were used as negative controls. Each assay was carried out three times and average values were obtained.

Lipid peroxidation

Lipid peroxidation levels of finger millet germinating seedlings were measured according to the method described by Heath and Packer (Heath and Packer 1968).

cDNA synthesis and cloning of lipoxygenase gene

Total RNA was extracted from 100 mg finger millet germinating seedlings with RNA Iso plus reagent (Takara Ostu, Shiga, Japan), following the manufacturers protocol. The first strand cDNA was synthesized by reverse transcription using 2 µg of

total RNA and Oligo (dT) 18 primer according to the manufacturer's instructions (Prime script 1st strand cDNA synthesis kit Takara Ostu, Shiga, Japan). One tenth volume of cDNA was used for LOX gene amplification using the LOX gene specific primers (Table 1) which were designed from highly conserved regions of *Poaceae* family LOXs. Confirming to the manufacturer's recommendation (Takara Ostu, Shiga, Japan), PCR reactions were carried out using Taq DNA polymerase in an Eppendorf Gradient Master-thermo cycler (Eppendorf, Hamburg, Germany) using the following PCR conditions in sequence; Denaturation at 94 °C for 2 min, 35 repetitive cycles of 94 °C for 1 min, 55 °C for 30 s, and 72 °C for 1 min and the reaction was completed with a final extension at 72 °C for 5 min. The amplified PCR products were analyzed using 1 % agarose gel electrophoresis to verify for size and to confirm gene specific amplification. The PCR products were purified using PCR prep DNA purification kit (Gene Jet DNA Extraction kit, Fermentas, Thermo Fisher Scientific Inc., Waltham, MA USA) and cloned into a pTZ57R/T vector (Fermentas, Thermo Fisher Scientific Inc., Waltham, MA USA). The recombinant plasmids were sequenced from both directions for confirmation. The 5' and 3' RACE PCR were conducted to obtain the missing 5' and 3' end sequences using the 5' and 3' RACE kit/2nd generation (Roche, Basel, Switzerland) following the manufacturer's instructions and conditions. The obtained RACE PCR fragments were separated using 1 % agarose gels, cloned and sequenced. Finally obtained *EcLOX* full-length cDNA was deposited in the GenBank (accession number: KF813007).

Sequence and phylogenetic analysis

Sequence analysis of *EcLOX* was accomplished using the NCBI-BLAST search service (Altschul et al. 1990). ExpASY translate tool (web.expasy.org/translate/) was used to deduce the amino acid sequence. Nucleotide and protein sequences of *EcLOX* were aligned with other LOX sequences using ClustalW (<http://www.ebi.ac.uk.clustalw>). The isoelectric

Table 1 Primers used for *EcLOX* cloning

Primer name	Primer sequence (Sense)
Ec F1	5'-CAGGCGTGGTGAAGGAG-3'
Ec F2	5'-CGCCAATGACACCTATCTGCCTAG-3'
5' RACE	5'-ATGTTCTGGCACGGGGTCG-3'
3' RACE	5'-ATGCTTCTGTTCCCAAC-3'
	Primer sequence (Antisense)
Ec R1	5'-GGAAATGCTGTTGGGATG-3'
Ec R2	5'-CTCCTTCCACCACGCCTG-3'
Ec R3	5'-GATCCGTCTATCCACAGTAACCGC-3'
3' RACE	5'-GACCACGCGTATCGATGTCGA CTTTTTTTTTTTTTTV-3'

point (pI) and molecular weight (MW) of protein were computed using the Compute pI/Mw Tool (http://web.expasy.org/compute_pi/). The proteins secondary structure was predicted using SOPMA (Self Optimized Prediction Method with Alignment). The topography of the *EcLOX* protein was investigated using Conserved Domain Architecture Retrieval Tool (CDART) software (<http://www.ncbi.nlm.nih.gov/Structure/lexington/lexington.cgi>) (Geer et al. 2002).

Phylogenetic analysis was performed to compare the protein sequence of *EcLOX* with other plant LOXs. The arachidonate 12-LOX amino acid sequence [*Mus musculus*, accession number: CAA74714] was included as an out-group control. The alignments were employed for phylogenetic analysis using MEGA 5 software (<http://www.megasoftware.net/>) by Neighbor-joining method and bootstrap consensus tree was inferred from the 1000 tree replicates.

Temporal, tissue specific and hormonal regulation studies

To analyze gene expression levels during seedling growth and development, finger millet seedlings at every 24 h interval for 7 days were collected for tissue specific expression. Four-day-old seedlings were selected and their leaves and roots were collected. For hormonal regulation studies on *EcLOX* gene, exogenous plant hormones like methyl jasmonate (MeJA) and salicylic acid (SA) at 100 μ M and 2.5 mM concentrations separately were dissolved in 10 % ethanol to test the influence of these hormones on 7 days old plants. The plants which received 10 % ethanol were considered as controls and plant samples were collected at different time intervals and stored in -80°C for further studies. The total RNA was isolated from above samples using the RNA Iso plus reagent (Takara Ostu, Shiga, Japan). 2 μ g of the isolated total RNA was used in reverse transcription reaction to synthesize cDNA using the Primescript 1st strand cDNA synthesis kit (Takara Ostu, Shiga, Japan) according to manufacturer's instructions. One tenth volume of cDNA was used as template in PCR reactions using LOX gene specific primers Ec F2: 5'-CGCCAATGACACCTATCTGCCTAG-3', Ec R2: 5'-CTCCTTCCACCACGCCTG-3' to analyze the levels of expression by semi-quantitative RT-PCR in an Eppendorf Gradient Master-thermo cycler (Eppendorf, Hamburg, Germany) using following PCR conditions; Initial denaturation at 94°C for 2 min, followed by 30 cycles of amplification (30 s at 94°C , 30 s at 55°C , and 1.5 min at 72°C), and a final extension at 72°C for 5 min. Actin was used as an endogenous control with actin gene specific primers, EcActin-F: 5'-GATTGCCCTCCTCCTCCTC-3' and EcActin-R: 5'-TTGATTATGGAGCGGGTGAT-3' PCR conditions as follows; initial denaturation at 94°C for 2 min, tracked by 30 cycles of amplification (94°C for 30 s, 57°C for 30 s, and 72°C for 1 min) and a final extension at 72°C

for 5 min. The amplified PCR products were analyzed by 1 % agarose gel electrophoresis.

Homology modeling and model validation

The protein sequence of *EcLOX* (887 amino acids) (accession number: AHH24969) was retrieved from NCBI (<http://www.ncbi.nlm.nih.gov/>) and the template for homology modeling was retrieved from the Protein Data Bank (PDB) (<http://www.rcsb.org/pdb/>) using BLAST (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) search. The structural template 2SBL having 839 amino acids (Boyington et al. 1997) was selected for homology modeling. Modeling was performed by using Modeller9v10 software. QUARK web server (<http://zhanglab.ccmb.med.umich.edu/QUARK/>), which predicts the protein structure by ab initio method, was used to predict the structure of N-terminal region (1–25 amino acid residues), as this region does not have similarity with the template selected for modeling. The final protein model was generated by joining the two protein models by multitemplate modeling, using Modeller9v10. The predicted homology model of *EcLOX* was evaluated using Structure Analysis and Verification Server (SAVES) (<http://nihserver.mbi.ucla.edu/SAVES/>), which harbors various analysis tools like PROCHECK (Laskowski et al. 1993), ERRAT (Colovos and Yeates 1993) and Verify3D (Eisenberg et al. 1997). Finally the Model was subjected to MD simulations, to check the stability and compactness of the protein Model.

Molecular dynamics (MD) simulations

To study the structural stability of predicted model, MD simulations were executed using GROMACS 5.0.4 software package, with OPLS force field and the flexible SPC water model. The protein was immersed in a cubic box of 1 nm, filled with water. The Particle Mesh Ewald method (Essmann et al. 1995) was used for the calculation of electrostatic energy. A cutoff distance of 1.0 nm was given for calculating coulombic and Van der Waal's interactions. Energy minimization was done using the steepest descent method for 1000 steps. Equilibrations were performed at 300 K and 1 bar pressure for 1000 ps. LINCS (Hess et al. 1997) constraints were applied for all bonds, making the protein molecule fixed and allowing the movement of water molecules pertaining to protein. The Berendsen pressure and temperature coupling was used for connecting the system to an external bath (Berendsen et al. 1984) and the MD simulations were performed for a time period of 10 ns.

Molecular docking studies

To gain an insight into the active site pocket of *EcLOX* enzyme, molecular docking studies were implemented using

Schrödinger maestro software (version 9.2; Schrödinger LLC, New York) and the predicted 3D model of *EcLOX*. The results were considered by the G Scores obtained from GLIDE (Grid-based Ligand Docking with Energetics) docking [Suite 2012: GLIDE, version 5.8, Schrödinger, LLC, New York, NY, 2012.].

Protein preparation

Prior to docking, proteins were prepared using protein preparation wizard. Hydrogen atoms were added by assigning charges. Subsequently, the model was optimized and energy minimized using OPLS2005 force field. The output of this wizard was submitted to Receptor grid generation tool.

Receptor grid generation

A grid was generated around the active site of the enzyme by specifying the key active site residues reported in the literature using Receptor grid generation tool of GLIDE. A receptor grid was generated around the protein active site by choosing the residues V390, S529, H532, L534, H537, H542, H576, L579, A580, L584, N732, Q735, T747, I792, and I887 described in the literature (Kuhn et al. 2002; Andreou and Feussner 2009; Feussner and Wasternack 2002; Gan et al. 1996; Hemak et al. 2002). Grid box size was set at $20 \text{ \AA} \times 20 \text{ \AA} \times 20 \text{ \AA}$ and Van der Waals radii of receptor atoms were scaled to 1.00 $\text{\AA}$ with a partial atomic charge of 0.25.

Ligand preparation

The ligands used as inputs for docking were designed using ChemBioOffice 2010 software and were prepared using LigPrep (version 2.5, Schrödinger Inc.). Preparation of ligands includes assigning bond orders, generating various tautomers, ring conformations and stereo chemistries. All the conformations generated were minimized using OPLS-AA force field before proceeding for docking studies.

Docking

Docking studies were performed to explore the possible binding modes of the substrates and available inhibitors in the active site of the *EcLOX*. GLIDE module of Schrödinger was used for the docking studies. All docking calculations were performed using Standard Precision (SP) mode. A scaling factor of 0.8 and a partial atomic charge of less than 0.15 were applied to the atoms of the protein. The poses generated per ligand were set to 10,000 and the pose chosen for energy minimization was set to 10. GLIDE docking score was used to determine the best docked structure from the output. The interactions of these docked complexes were studied further.

Results and discussion

Lipoxygenase activity and lipid peroxidation levels in finger millet germinating seedling

The change in LOX enzyme activity of finger millet was tested from day 1 to day 8 of development after seed germination. The enzyme activity was initially examined at different pH conditions (pH 4–9) and found that the *EcLOX* activity has an optimum pH of about 6.4 (data not shown). After seed germination the activity levels were found to be increasing from day 1 and reached peak on day 4, there after declined gradually during development (Supplementary Fig. 1A).

The life cycle of seed plant begins with imbibition and germination. During the life cycle of plants several

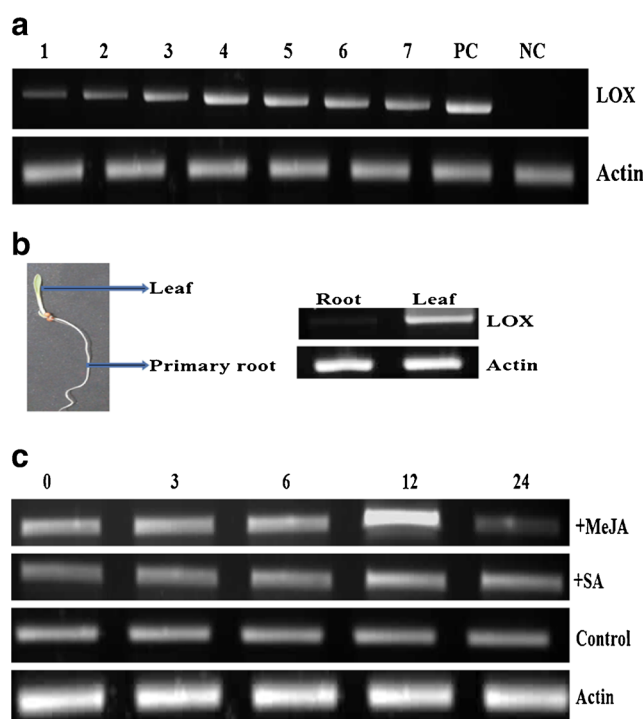


Fig. 1 a) Developmental expression levels of *EcLOX* transcript during seed germination. Total RNA was isolated from finger millet seedling at every 24 h' time interval after germination for 7 days and utilized to measure the levels of LOX gene expression using gene specific primers. Cloned LOX plasmid was used as the internal positive control (PC) and tube without addition of cDNA was used as the negative control (NC) in the semi-quantitative PCR. Actin specific primers were used to amplify actin gene as the control gene (endogenous control). b) Semi-quantitative RT-PCR analysis of organ specific *EcLOX* transcript. Total RNA was extracted from the leaf and root of finger millet germinating seedlings. Equal loading of RNA was confirmed by expression of Actin (endogenous control). c) Semi-quantitative RT-PCR analysis of *EcLOX* in response to Methyl Jasmonate (MeJA (100 mM) and SA (2.5 mM). 7th day old germinating seedlings were selected for hormonal treatment and samples were harvested at 0, 3, 6, 12 and 24 h post treatment. Plants received ethanol were (–MeJA, –SA) served as control. Actin specific primers were used to amplify actin gene as an additional control (endogenous control) in semi-quantitative RT-PCR

developmental alterations occur characterized by changes in the composition of intracellular lipids which serve as a major carbon source for growth of seedlings (Gerhardt et al. 2005). It is believed that, during germination, oxygenases play an active role in oxidation of PUFAs, which are then converted into sugars via gluconeogenesis to provide further metabolic energy in the cell. Recent reports using proteome profiling identified that LOXs are involved in lipid degradation, protease and 26S proteasome systems are involved in protein degradation in soybean and rice seedlings (Han et al. 2013). Increased LOX activity levels was observed in many plants during germination (Feussner et al. 2001). The lipid peroxidation (LPO) level in finger millet seedlings increased gradually from the day of germination, demonstrating higher levels on day 5, and declined thereafter (Supplementary Fig. 1B). It was suggested that the lipid peroxidation and its propagation by LOXs are essential for cell elongation after germination (Rao et al. 1998). Free radicals form continuously in cells as accidental by-products of metabolism or deliberately by enzymatic reactions. Metal ions like iron can induce lipid peroxidation via non-radical mediated mechanisms. Enhanced lipid peroxidation levels in our studies coincided with a high LOX activity which may result in AOS production; high level of LOX activity comes prior to an insignificant increase in AOS production in leaves of *Solanum* (Polkowska-Kowalczyk et al. 2004). LOXs play a significant role in the metabolism of membrane lipids to lipid peroxides through a free radical mediated enzymatic reaction.

Cloning of full-length *EcLOX* gene and sequence analysis

The gene specific primers were used for PCR amplification reactions to clone LOX gene from finger millet germinating seedlings, these primers were designed from the conserved regions of the known LOX gene sequences of *poaceae* family (Table 1). The amplified products with expected length were extracted from the agarose gel, cloned and the recombinant plasmids were sequenced from both ends for confirmation. A partial gene fragment with 2569 bp was obtained without 5' and 3' end regions. The missing 5' and 3' ends for LOX gene were acquired by 5' and 3' RACE method by designing primers from the partial sequence of LOX gene fragment. The obtained full length sequence of LOX cDNA was found to have 2940 bp and contains an open reading frame (ORF) of 2661 bp that encodes 887 amino acid residues with approximate molecular weight of 100.80 kDa, with a calculated isoelectric point of 5.90. The BLAST search of this full length cDNA revealed significant similarity with other plant LOXs. Therefore, it was designated as *EcLOX*. A BLAST homology search revealed that the deduced amino acid sequence of *EcLOX* shared high identity (85 %) to *Setaria italica* (accession number: XP_004982082.1) LOX protein, which belongs to the *poaceae* family. *EcLOX* also shared high resemblance to other plant LOX

proteins, presenting 83 %, 76 %, 73 % and 71 % identity with *Zea mays* (accession number: XP_003627185), *Oryza sativa* (accession number: ABD47523.1), *Triticum aestivum* (accession number: ADZ31265.1) and *Hordeum vulgare* (accession number: AAB60715.1). Multiple sequence alignment showed that *EcLOX* contained conserved regions which are well known in plant LOXs (Siedow 1991). The topography of the *EcLOX* protein was investigated using Conserved Domain Architecture Retrieval Tool (CDART) software (<http://www.ncbi.nlm.nih.gov/Structure/lexington/lexington.cgi>). The results of this study showed that *EcLOX* proteins consisted of two domains; Domain I from amino acids 38–165, namely a PLAT (Polycystin-1, Lipoxygenase, Alpha-Toxin) domain in the amino-terminal region and domain II from amino acids 168–887, a LOX domain. The generally accepted function of the PLAT domain is to mediate interactions with lipids or membrane-bound proteins. Sequence alignment of *EcLOX* with other plant LOXs showed that the deduced amino acid sequence harbored all three conserved His residues (537, 542, and 728), Asn732 and Ile887, which have been shown to be crucial for binding of LOXs to iron atom in the active site (Supplementary Fig. 2). Previous studies identified a conserved Thr/Val motif that may determine 9-LOX regio-specific activity (Hornung et al. 1999). The amino acid sequence of *EcLOX* contained the Val (V) residue at position 595, suggesting that *EcLOX* belongs to 9-LOX subfamily. In addition, the conserved Arg (R745) residue which interacts with the carboxyl group of fatty acids is the determining factor for the inverse substrate orientation in plant LOXs, and the Ala (A580) residue that determines the S-stereospecificity of most LOXs is also present in *EcLOX* protein (Padilla et al. 2012) (Supplementary Fig. 2).

The secondary structure prediction by SOPMA server (Geourjon and Deleage 1995) indicated that deduced *EcLOX* contained 36.64 % of α -helices, 10.15 % of β -turns, 14.99 % of extended strands, and 38.22 % of random coils, respectively. The sub cellular localization of *EcLOX* was predicted by utilizing different publicly available programs, Plant-mPLoc, EpiLoc, YLoc, ProtComp (<http://linux1.softberry.com>), TargetP (<http://www.cbs.dtu.dk/services/TargetP/>), ChloroP (<http://www.cbs.dtu.dk/services/ChloroP/>) and WolfPsort (<http://wolfsort.org>). These programs collectively predicted that *EcLOX* is most likely localized in cytoplasm.

Phylogenetic analysis

To decide the phylogenetic relationship between *EcLOX* and members of other plant LOXs, full-length protein sequences were retrieved from GenBank and sequence alignment was performed followed by phylogenetic tree construction. On observation, the phylogenetic tree was found to be bifurcated into four distinct clades which include Dicot 9-LOXs, Dicot 13-LOXs, Monocot 9-LOXs, and Chloroplastic LOXs supported by good bootstrap values (Supplementary Fig. 3). *EcLOX* was found to

be grouped with monocot 9-LOXs. The 9-LOXs gene group had valine in the conserved motif. *ZmLOX*, and *OsLOX* (accession numbers: ABC59688 and ABD47523, respectively) were found to be grouped with *EcLOX*, which suggests that they are orthologous.

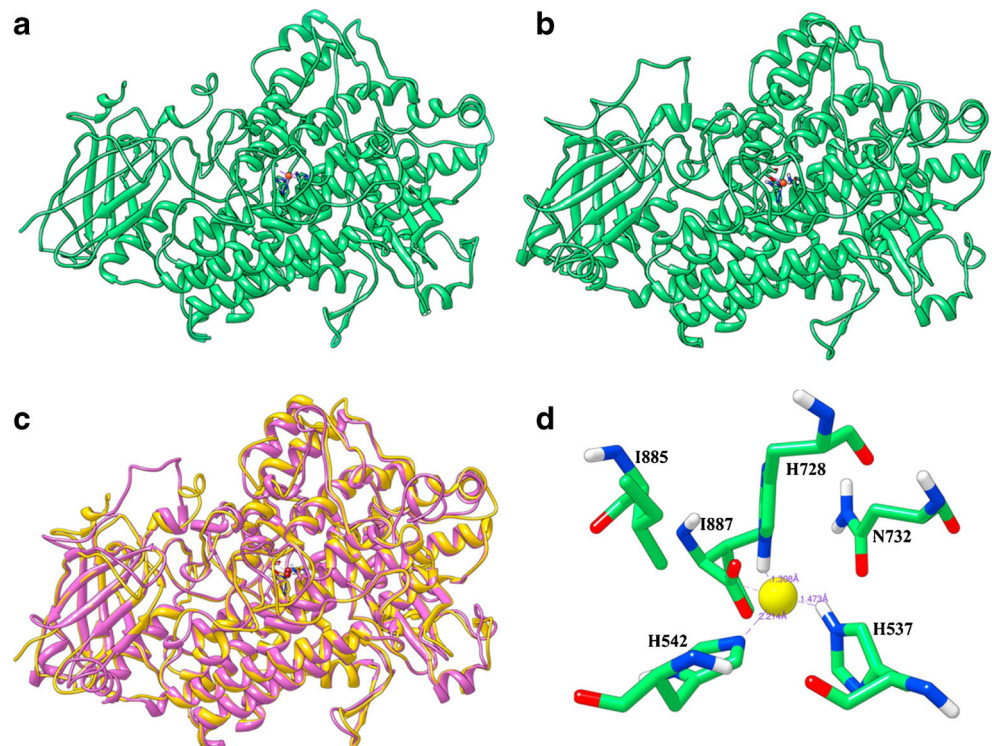
Temporal, tissue specific and hormonal regulation studies of finger millet seedling lipoxygenase

Semi-quantitative RT-PCR was used to investigate the developmental changes and the distribution of *EcLOX* gene in finger millet growing plant tissue. Mobilization of storage lipids is an essential process at the time of germination. During early stages of germination, a lipid body dependent LOX was observed which involves in oxygenation of lipid bodies, the first step in mobilization of storage lipids for providing energy and carbon equivalents (Porta and Rocha-Sosa 2002). High levels of finger millet seedling LOX gene expression was detected on day 4 and day 5 of germination which dropped gradually afterwards on day 6 and 7 of germination (Fig. 1a). New LOXs were synthesized in the seedling, during germination and after germination and within few hours to few days leads to extreme levels of accumulation of LOX protein. The LOX mRNAs synthesized during germination could also be found in the mature plant (Feussner et al. 2001). In tissue specific analysis the highest levels of expression was observed in leaf and lowest expression was found in root (Fig. 1b). Such differential distributions of *EcLOX* transcript in diverse plant

tissues suggest the different physiological roles during plant growth and development. The isoforms of specific LOX expression has been reported in other plants species in association with meristematic activity of young tissues. A specific 9-LOX appears to be crucial for tuber development in potato (Kolomiets et al. 2001). During the seed germination high level of 9-LOX expression was observed but the biological function of the 9-LOX is unclear. The 9-LOX expression may be induced during embryo development or cell division process by the synthesis of some signaling or hormonal compounds that include products like hexenal, traumatin and jasmonates. 9-LOXs were possibly implicated in providing an energy source for seed germination by oxidation of storage lipids into substrates (Mizuno et al. 2003).

The hormones MeJA and SA function as signaling molecules in plants with response to abiotic stress or defense in plants (Glazebrook et al. 2003). MeJA, is an important plant defense response signaling molecule in the regulation of defense responses to herbivory (Howe and Jander 2008) and SA is an important signaling molecule present in plants, which induces tolerance to stress and plays an important role in plant physiological processes such as effect on growth and development, membrane permeability, ion up take and transport (Horváth et al. 2007). Finger millet 7th day germinating seedlings were selected where LOX gene expression was not detected and seedlings were treated with 100 μ M MeJA, and 2.5 mM SA; the obtained results clearly indicated that exogenous MeJA induced an

Fig. 2 Results of *in silico* structural characterization of *EcLOX* protein **a**) 3-dimensional structure of the template (PDBID: 2SBL) represented as green ribbon. **b**) The predicted model of *EcLOX*. **c**) Superimposition of the 3-dimensional model of *EcLOX* (pink) compared with the template (PDBID: 2SBL) (yellow). Metal binding amino acid side chains are shown in stick representation. Iron binding to the residues is shown in brown color. **d**) The amino acid residues involved in interaction with the Iron (Fe^{2+}) were shown as sticks interacting with Iron (yellow). The coordinate bonds were represented as blue lines and the respective bond lengths were mentioned



expression of the *EcLOX* gene (Fig. 1c). Maximum expression of *EcLOX* after the MeJA treatment was at 12 h and decreased thereafter, unlike MeJA, exogenous SA did not induce the *EcLOX* mRNA accumulation; no noticeable induction was observed in response to SA (Fig. 1c). Thus *EcLOX* appears to be up-regulated by MeJA but not by SA. Similar kind of expression of LOX gene was observed in *Camellia sinensis* (L.) and (*CsLOX1*) (Liu and Han 2010), in *Caragana jubata* (*CjLOX*). Strong up-regulation was observed at 12 h exposure to MeJA (Bhardwaj et al. 2011). The LOX mRNAs synthesized during germination could also be found in the mature plant, the levels of LOX transcript increases by applying hormones like ABA, JA or by environmental or physiological stress such as water deficit, wounding and biological stress like pathogen attack (Feussner et al. 2001).

Homology modeling and MD simulations

The protein sequence of *EcLOX* was investigated against the Protein Data Bank (PDB) using BLASTp server, which retrieved three protein structures (PDB ID's: 1IK3, 1ROV, 2SBL) as templates with highest identity of 53 % and query coverage of 97 %. After verifying the quality of the structures, the one with no mismatches (PDB ID: 2SBL) was selected as the template for homology modeling (Fig. 2a). Due to the lack of sequence similarity between target and template sequences for first 25 amino acid residues, this part of the target sequence was modeled using QUARK online web server. Homology modeling was performed by multiple modeling methods using the best model predicted by QUARK and the protein structure of 2SBL as templates. Totally ten homology models were generated and among them model 5 was selected as the best

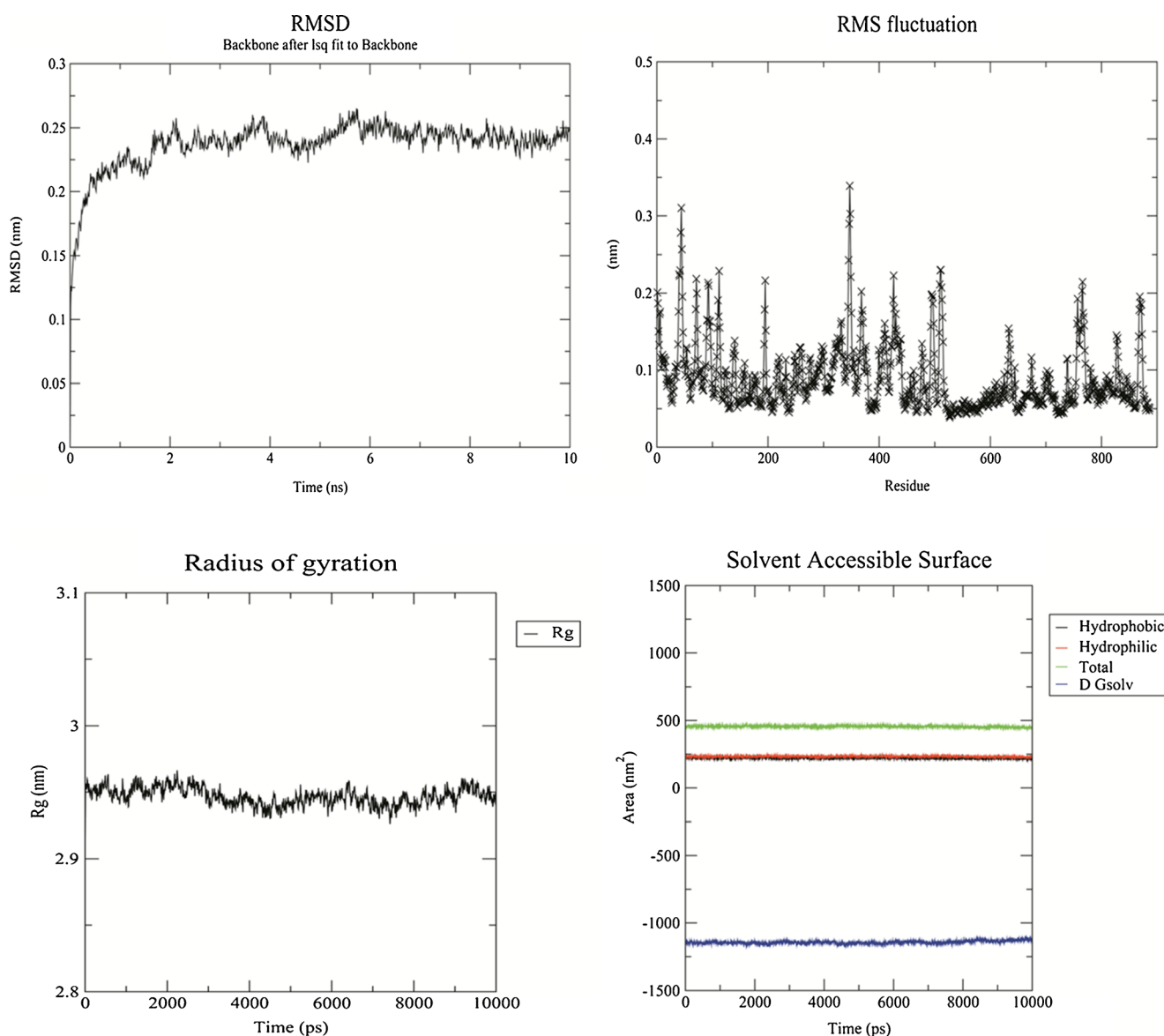


Fig. 3 Depiction of the MD simulations results of predicted *EcLOX* protein performed for a period 10 ns

model, as it has the least value of molPDF score (6542.31396) and a lower DOPE score (−105056.57813) value. The molPDF and DOPE scores of all the models generated were provided in Supplementary Table 1. The quality of the model generated was validated by submitting to SAVES server. The stereo chemical quality of the model was checked using PROCHECK, which revealed that 90.9 % of the residues were in the most favored regions (Supplementary Fig. 4A). Verify3D analysis disclosed that 94.26 % of the residues had an average 3D-1D score > 0.2, which shows that the model is reliable (Supplementary Fig. 4B). The overall quality factor of the protein was also found to be good (89.867). The ProSA

(Wiederstein and Sippl 2007) score of −11.98 for the model, which is closer to that of template score (−12.57), indicates that the energetic architecture of the predicted model was good (Fig. 2b). Superposing of template and target structures revealed that the metal binding residues (His537, His542, His728, Asn732, and Ile887) were conserved besides forming interactions with iron ion (Fig. 2c, d). The modeled protein have two domains namely an N-terminal PLAT domain with an eight-stranded antiparallel β -barrel topology and a C-terminal catalytic domain, which comprises of a bundle of 37 helices having large central helix. The Fe^{2+} ion is coordinated to four amino acids at the center of the catalytic domain.

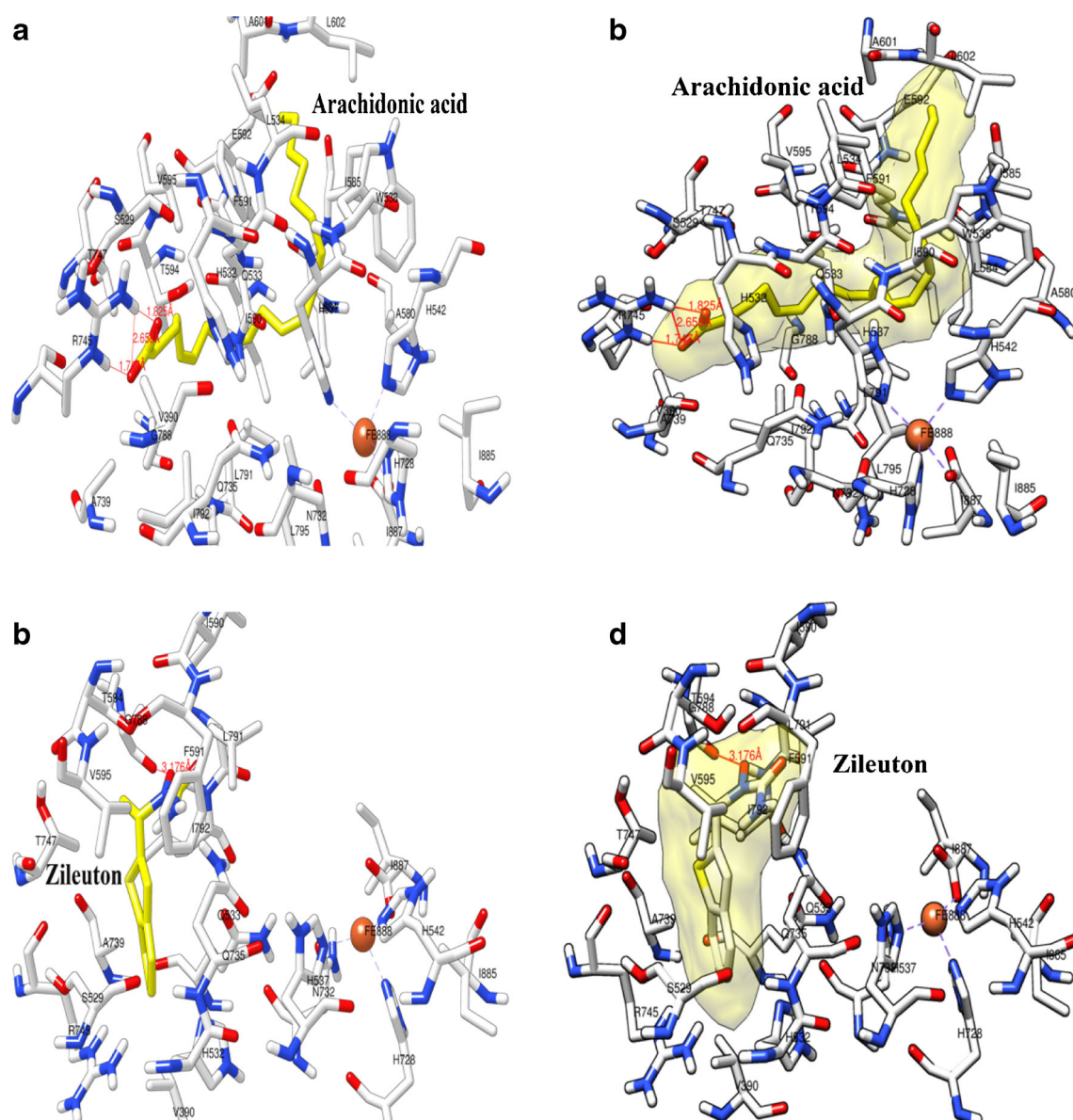


Fig. 4 Predicted ligand-binding modes of **a**) Arachidonic acid and **b**) Zileuton in the active site of *EcLOX*. **c** and **d**) Surface view of Arachidonic acid and Zileuton respectively, in the active site of *EcLOX*. The atoms were represented as sticks. Iron atom was colored brown. Carbon, Oxygen and Nitrogen atoms of protein were represented in

white, red, and blue colors, respectively. The carbon atoms of the ligands were represented in yellow colour. The hydrogen bonds were represented as red lines and labeled with distances. Surface of the compounds was shown in yellow colour

Table 2 Docking results of *EcLOX* with its natural substrates

Natural substrates of lipoxygenase				
Ligand	XP Glide Score Kcal/mol	Glide Energy Kcal/mol	H-bond Num	Interacting residues
Arachidonic acid CID: 444899	-6.718	-26.524	3	V390, S529, H532, L534, H537, W538, H542, A580, L584, I585, F591, E592, T594, V595, A601, L602, H728, N732, Q735, A739, R745, T747, L791, I792, L795, I885, I887
Linoleic acid CID: 5280450	-4.041	-35.790	3	V390, S529, H532, L534, H537, W538, H542, A580, L584, I585, F591, V595, T594, A601, L602, H728, N732, Q735, A739, R745, T747, L791, L795, I885, I887
Linolenic acid CID: 5280934	-3.583	-33.264	3	V390, S529, H532, L534, H537, W538, H542, L584, I585, I590, F591, T594, V595, H728, N732, Q735, A739, R745, T747, 788, L791, I792, I885, I887

The predicted protein model of *EcLOX* was subjected to MD simulations for a time period of 10 ns to gain knowledge about the conformational flexibility and structural stability. To achieve this, Root mean square deviations (RMSD), Root mean square fluctuations (RMSF), and Radius of gyration (Rg) of the protein during the simulations were monitored. The RMSD profiles calculated from the starting structure displayed that the protein was stable and showed less deviation (less than 0.3 nm) throughout the 10 ns simulation time period. The RMSF calculated for the trajectories indicate that majority of the residues were stable with fluctuations not more than 0.2 nm. The Rg calculated over the trajectory showed a decrease in the value, which implies that the compactness of the protein increased. The overall fluctuation in Rg from 2.91 nm to 2.95 nm shows that the conformational changes were less during the simulation. Solvent Accessible Surface Area (SASA) analysis was performed to identify the changes occurred on the protein surface during the 10 ns simulation time. SASA analysis helps in geometric measurement of the magnitude of an amino acid to interact with its surroundings. This interaction is proportional to accessibility of an amino acid to its environment (Shaytan et al. 2009; Lins et al. 2003). Variations in SASA levels infer that the change in

degree of amino acid exposure, affects the proteins structure. The graph infers that no significant changes occurred in SASA and had maintained almost constantly throughout the simulation time period, with suggests that the hydrophobic protein core is maintaining its compactness. The above structural analysis strongly expresses that the modeled protein structure is good and reliable. The RMSD, RMSF, Rg and SASA graphs were represented in Fig. 3. The protein model with least potential energy from the 10 ns simulation trajectory was extracted and used for further studies.

Molecular docking studies

LOXs participate in various physiological roles in both plants and animals. The major role of LOXs include membrane lipid peroxidation. Hyperactivity of LOXs may increase the levels of conjugated unsaturated fatty acids and hydroperoxides in living tissues, which may be responsible for various adverse effects in plants (Baraniak and Krzepilko 2004). Inhibition of LOXs helps in controlling various diseases like asthma, atherosclerosis and inflammatory diseases in animals (Singh et al. 2004). Owing to their therapeutic potential, understanding the properties of the active site to find efficient inhibitors

Table 3 Docking results of *EcLOX* with inhibitors

Approved inhibitors of lipoxygenase				
Ligand	XP Glide Score Kcal/mol	Glide Energy Kcal/mol	H-bond Num	Interacting residues
NDGA CID: 4534	-6.129	-45.810	5	N148, I146, Y205, E269, D268, F271, H537, H542, P568, S567, Y570, R571, H728, N732, D809, P808, E810, W814, I885, I887
Diethyl Carbamazepine CID: 3052	-5.417	-17.805	0	Q533, L534, H537, W538, H542, I576, A580, L584, I585, I590, F591, A601, L602, H728, N732, L791, L795, I885, I887
Zileuton CID: 60490	-5.119	-17.258	1	V390, S529, H532, H537, Q533, H542, I590, F591, T594, V595, H728, N732, Q735, A739, T747, L791, I792, I885, I887

is essential. Observation of the enzyme-substrate binding modes discloses the possible binding interactions and provides structural insights. So, this study was an attempt made to explore the binding modes of substrates and available inhibitors in the active site of *EcLOX*. The natural substrates of LOXs used for docking are Arachidonic acid, Linoleic acid and Linolenic acid, whereas the reported inhibitors used for docking are Nordihydroguaiaretic acid (NDGA), Diethyl Carbamazone, and Zileuton (Carter et al. 1991; Bozza et al. 1996). Docking studies showed that all the substrates were appropriately positioned into a deep U-shaped tunnel like catalytic site adjacent to metal ion. The tunnel is surrounded by large number of hydrophobic residues which interact with each other to form compact hydrophobic clusters. The carboxyl group of all the substrates was found to be forming three hydrogen bond interactions with the guanidine group of R745 present at the opening of the pocket, which implies that this interaction is very much essential for the enzyme action. Similar type of interaction was also reported in an earlier study on human 15-LOX, which stated that a positively charged amino acid residue at the opening of the cavity is essential for the appropriate positioning of carbon atoms of the fatty acid chain in the active site (Gan et al. 1996). The orientation of fatty acid chain in the active site of *EcLOX* was found to be inversed to that one present in the animal LOXs, supporting the earlier reports (Padilla et al. 2012). The residues lining walls of the catalytic site include V390, S529, H532, Q533, L534, H537, W538, H542, A580, L584, I585, F591, T594, V595, A601, L602, H728, N732, A739, R745, T747, L791, I792, L795, I885, and I887, most of which are hydrophobic in nature. Two polar amino acid residues E592 and Q735 were observed adjacent to the iron center. Similar observation was also made in an earlier study on rabbit reticulocyte 15-LOX (Borngräber et al. 1999). Investigation of binding modes of inhibitors revealed that the inhibitors were sharing similar interactions as substrates, blocking the path of tunnel. This infers that the inhibitors may act as competitors for substrates in binding to the active site. The high docking score of NDGA compared to other inhibitors might be due to the large number of hydrogen bonds formed by it in the active site. The binding modes of Arachidonic acid and Zileuton were characterized in Fig. 4. The binding modes of other compounds were given as supplementary Figs. 5–8. The list of compounds used for the docking with corresponding GLIDE scores, GLIDE energies and the interacting amino acid residues identified were shown in Tables 2 and 3.

A full length cDNA (*EcLOX*) was cloned from finger millet germinating seedlings. This transcript showed high expression on day 4 after germination and in tender leaf, whereas low level of expression was observed in root. The LOX enzyme activity and lipid peroxidation levels were more on day 4 and day 5 after seed germination. This *EcLOX* expression could be induced by MeJA and no effect was observed on inducing

with SA. Phylogenetic analysis revealed that the *EcLOX* belongs to type-I, 9-LOX family. Structural characterization of this deduced protein (*EcLOX*) sequence showed that the constructed 3-dimensional model could be considered as a working model for generating hypotheses, designing further experimental studies, and useful for more accurate prediction of LOX functions and binding affinities of inhibitors. However, further characterization is essential to confirm it. The obtained results may serve as a very important preliminary data for further characterization of *EcLOX* gene and its corresponding isoenzymes role in patho-physiology of millets.

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Conflict of interest The authors declare that they have no conflict of interest.

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